# C Project 🏨

```
#include <stdio.h>
#include <string.h>

#define MAX_SEATS 2

struct Hostel
{
    int hostelNo;
    char name[50];
    char gender[10];
    char block;
    char plot[20];
    char flat[20];
};

struct Hostel hostels[] =
{
    {101, "Bashundhara Girls Hostel 1", "Girl", 'A', "Plot-12", "Flat-4A"},
    {102, "Bashundhara Girls Hostel 2", "Girl", 'B', "Plot-15", "Flat-3B"},
    {103, "Bashundhara Girls Hostel 3", "Girl", 'C', "Plot-18", "Flat-5C"},
    {104, "Bashundhara Girls Hostel 4", "Girl", 'D', "Plot-22", "Flat-2D"},

    {201, "Bashundhara Boys Hostel 1", "Boy", 'A', "Plot-35", "Flat-3A"},
    {202, "Bashundhara Boys Hostel 2", "Boy", 'B', "Plot-39", "Flat-5B"},
    {203, "Bashundhara Boys Hostel 3", "Boy", 'C', "Plot-43", "Flat-2C"},
    {204, "Bashundhara Boys Hostel 4", "Boy", 'D', "Plot-47", "Flat-6D"}
};


/* Count bookings for a hostel */
int getBookedSeats(int hostelNo)
{
    FILE *file;
    char line[200];

    int bookingID;
```

```
    int bookedHostel;
    int count = 0;

    file = fopen("bookings.txt", "r");

    if (file == NULL)
    {
        return 0;
    }

    while (fgets(line, sizeof(line), file) != NULL)
    {
        if (sscanf(line, "%d|%d", &bookingID, &bookedHostel) == 2)
        {
            if (bookedHostel == hostelNo)
            {
                count++;
            }
        }
    }

    fclose(file);

    return count;
}


/* Generate next booking ID */
int getNextBookingID()
{
    FILE *file;
    char line[100];

    int bookingID;
    int hostelNo;
    int nextID = 1001;
```

```
    file = fopen("bookings.txt", "r");

    if (file == NULL)
    {
        return nextID;
    }

    while (fgets(line, sizeof(line), file) != NULL)
    {
        if (sscanf(line, "%d|%d", &bookingID, &hostelNo) == 2)
        {
            if (bookingID >= nextID)
            {
                nextID = bookingID + 1;
            }
        }
    }

    fclose(file);

    return nextID;
}


/* Save booking information */
int saveBooking(int bookingID,
                int hostelNo,
                char name[],
                char phone[],
                char roomType[],
                char furniture[],
                char food[],
                int totalCost)
{
    FILE *file;

    file = fopen("bookings.txt", "a");
```

```
    if (file == NULL)
    {
        printf("\nError: Could not save booking.\n");
        return 0;
    }

    fprintf(file, "%d|%d|%s|%s|%s|%s|%s|%d\n",
            bookingID,
            hostelNo,
            name,
            phone,
            roomType,
            furniture,
            food,
            totalCost);

    fclose(file);

    return 1;
}


int main()
{
    int genderChoice;
    int hostelChoice;
    int hostelIndex = -1;

    int roomChoice;
    int furnitureChoice;
    int foodChoice;

    int bookChoice;
    int confirm;

    int roomPrice;
```

```
    int furnitureCost;
    int foodCost;
    int blockCost;
    int totalCost;

    int bookedSeats;
    int remainingSeats;

    int i;

    char gender[10];
    char roomType[30];
    char furniture[20];
    char food[20];

    char name[50];
    char phone[20];

    int bookingID;



printf("\n=============================================\n");
    printf("      BASHUNDHARA HOSTEL FINDER\n");
    printf("          & BOOKING SYSTEM\n");
    printf("=============================================\n");


    /* Select gender */

    printf("\nAre you looking for a hostel for?\n");
    printf("1. Boy\n");
    printf("2. Girl\n");

    printf("\nEnter choice: ");
    scanf("%d", &genderChoice);
```

```
if (genderChoice == 1)
{
   strcpy(gender, "Boy");
}
else if (genderChoice == 2)
{
   strcpy(gender, "Girl");
}
else
{
   printf("\nInvalid choice!\n");
   return 0;
}


/* Show available hostels */

printf("\n=========================================\n");
printf("       %s HOSTELS\n", gender);
printf("=========================================\n");

for (i = 0; i < 8; i++)
{
   if (strcmp(hostels[i].gender, gender) == 0)
   {
      bookedSeats = getBookedSeats(hostels[i].hostelNo);
      remainingSeats = MAX_SEATS - bookedSeats;

      if (remainingSeats > 0)
      {
         printf("\nHostel No. : %d", hostels[i].hostelNo);
         printf("\nName       : %s", hostels[i].name);
         printf("\nBlock      : %c", hostels[i].block);
         printf("\nPlot       : %s", hostels[i].plot);
         printf("\nFlat       : %s", hostels[i].flat);
         printf("\nSeats Left : %d/%d\n",
               remainingSeats, MAX_SEATS);
```

```
        }
    }
}


/* Select hostel */

printf("\nEnter hostel number: ");
scanf("%d", &hostelChoice);

for (i = 0; i < 8; i++)
{
    if (hostels[i].hostelNo == hostelChoice &&
        strcmp(hostels[i].gender, gender) == 0)
    {
        hostelIndex = i;
        break;
    }
}

if (hostelIndex == -1)
{
    printf("\nInvalid hostel selection!\n");
    return 0;
}


/* Check availability */

bookedSeats = getBookedSeats(hostels[hostelIndex].hostelNo);

if (bookedSeats >= MAX_SEATS)
{

printf("\n=========================================\n");
    printf("            HOSTEL FULL\n");
```

```
printf("=========================================\n");

        printf("Sorry! %s already has %d/%d students.\n",
               hostels[hostelIndex].name,
               bookedSeats,
               MAX_SEATS);

        return 0;
    }

    remainingSeats = MAX_SEATS - bookedSeats;


printf("\n=========================================\n");
    printf("       HOSTEL AVAILABILITY\n");
    printf("=========================================\n");

    printf("Total Seats     : %d\n", MAX_SEATS);
    printf("Booked Seats    : %d\n", bookedSeats);
    printf("Remaining Seats : %d\n", remainingSeats);


    /* Room type */

    printf("\n=======================================\n");
    printf("             ROOM TYPE\n");
    printf("=======================================\n");

    printf("1. Master Bed - 10000 Tk\n");
    printf("2. Semi-Master Bed - 8000 Tk\n");
    printf("3. Partition Room - 5000 Tk\n");

    printf("\nEnter choice: ");
    scanf("%d", &roomChoice);

    if (roomChoice == 1)
```

```
{
    strcpy(roomType, "Master Bed");
    roomPrice = 10000;
}
else if (roomChoice == 2)
{
    strcpy(roomType, "Semi-Master Bed");
    roomPrice = 8000;
}
else if (roomChoice == 3)
{
    strcpy(roomType, "Partition Room");
    roomPrice = 5000;
}
else
{
    printf("\nInvalid room type!\n");
    return 0;
}


/* Furniture */

printf("\n=======================================\n");
printf("            FURNITURE\n");
printf("=======================================\n");

printf("1. Furnished (+1000 Tk)\n");
printf("2. Non-Furnished (+0 Tk)\n");

printf("\nEnter choice: ");
scanf("%d", &furnitureChoice);

if (furnitureChoice == 1)
{
    strcpy(furniture, "Furnished");
    furnitureCost = 1000;
```

```
}
else if (furnitureChoice == 2)
{
    strcpy(furniture, "Non-Furnished");
    furnitureCost = 0;
}
else
{
    printf("\nInvalid choice!\n");
    return 0;
}


/* Food */

printf("\n=======================================\n");
printf("          FOOD FACILITY\n");
printf("=======================================\n");

printf("1. With Food (+5000 Tk)\n");
printf("2. Without Food (+0 Tk)\n");

printf("\nEnter choice: ");
scanf("%d", &foodChoice);

if (foodChoice == 1)
{
    strcpy(food, "With Food");
    foodCost = 5000;
}
else if (foodChoice == 2)
{
    strcpy(food, "Without Food");
    foodCost = 0;
}
else
{
```

```
        printf("\nInvalid choice!\n");
        return 0;
    }


    /* Block charge */

    if (hostels[hostelIndex].block == 'A' ||
        hostels[hostelIndex].block == 'B')
    {
        blockCost = 1000;
    }
    else
    {
        blockCost = 500;
    }


    /* Total cost */

    totalCost = roomPrice + furnitureCost + foodCost + blockCost;


    /* Show details */


printf("\n=============================================\n");
    printf("            HOSTEL DETAILS\n");
    printf("=============================================\n");

    printf("Hostel No.  : %d\n", hostels[hostelIndex].hostelNo);
    printf("Hostel Name : %s\n", hostels[hostelIndex].name);
    printf("Block       : %c\n", hostels[hostelIndex].block);
    printf("Plot No.    : %s\n", hostels[hostelIndex].plot);
    printf("Flat No.    : %s\n", hostels[hostelIndex].flat);

    printf("---------------------------------------\n");
```

```
printf("Room Type : %s\n", roomType);
printf("Furniture : %s\n", furniture);
printf("Food      : %s\n", food);

printf("-----------------------------------------\n");

printf("Room Price   : %d Tk\n", roomPrice);
printf("Furniture    : +%d Tk\n", furnitureCost);
printf("Food         : +%d Tk\n", foodCost);
printf("Block Charge : +%d Tk\n", blockCost);

printf("-----------------------------------------\n");

printf("TOTAL COST : %d Tk/month\n", totalCost);

printf("=========================================\n");


/* Booking decision */

printf("\nWould you like to book this hostel?\n");
printf("1. Yes\n");
printf("2. No\n");

printf("\nEnter choice: ");
scanf("%d", &bookChoice);

if (bookChoice == 2)
{
    printf("\nNever mind, thanks for trusting us!\n");
    return 0;
}

if (bookChoice != 1)
{
    printf("\nInvalid choice!\n");
```

```
    return 0;
}


/* Student information */

getchar();

printf("\n========================================\n");
printf("          BOOKING INFORMATION\n");
printf("========================================\n");

printf("Enter your name: ");
fgets(name, sizeof(name), stdin);

name[strcspn(name, "\n")] = '\0';

printf("Enter your phone number: ");
scanf("%19s", phone);


/* Booking details */


printf("\n============================================\n");
printf("             BOOKING DETAILS\n");
printf("============================================\n");

printf("Name         : %s\n", name);
printf("Phone        : %s\n", phone);
printf("Hostel       : %s\n", hostels[hostelIndex].name);
printf("Hostel No.   : %d\n", hostels[hostelIndex].hostelNo);
printf("Block        : %c\n", hostels[hostelIndex].block);
printf("Plot No.     : %s\n", hostels[hostelIndex].plot);
printf("Flat No.     : %s\n", hostels[hostelIndex].flat);
printf("Room Type    : %s\n", roomType);
printf("Furniture    : %s\n", furniture);
```

```
    printf("Food         : %s\n", food);
    printf("Monthly Cost : %d Tk\n", totalCost);

    printf("=============================================\n");


    /* Final confirmation */

    printf("\nAre you sure you want to confirm this booking?\n");
    printf("1. Confirm Booking\n");
    printf("2. Cancel Booking\n");

    printf("\nEnter choice: ");
    scanf("%d", &confirm);

    if (confirm == 1)
    {
        bookingID = getNextBookingID();

        if (saveBooking(bookingID,
                        hostels[hostelIndex].hostelNo,
                        name,
                        phone,
                        roomType,
                        furniture,
                        food,
                        totalCost))
        {

printf("\n=============================================\n");
            printf("          BOOKING CONFIRMED!\n");

printf("=============================================\n");

            printf("Booking ID    : %d\n", bookingID);
            printf("Name          : %s\n", name);
            printf("Phone Number  : %s\n", phone);
```

```
            printf("Hostel       : %s\n", hostels[hostelIndex].name);
            printf("Hostel No.    : %d\n", hostels[hostelIndex].hostelNo);
            printf("Block        : %c\n", hostels[hostelIndex].block);
            printf("Room          : %s\n", roomType);
            printf("Furniture     : %s\n", furniture);
            printf("Food         : %s\n", food);
            printf("Monthly Cost  : %d Tk\n", totalCost);

            printf("----------------------------------------\n");
            printf("Thank you for choosing our hostel!\n");
            printf("We look forward to having you with us.\n");

printf("========================================\n");
        }
        else
        {
            printf("\nBooking was not confirmed because it could not be
saved.\n");
        }
    }
    else if (confirm == 2)
    {
        printf("\nBooking cancelled.\n");
    }
    else
    {
        printf("\nInvalid choice!\n");
    }

    return 0;
}
```